PYTHON BASICS — ONE-PAGE CHEAT SHEET

A quick reference for beginners | Read the concept, then run the example in VS Code

1. Variables & basic data types

Variables store values. Python decides the type automatically.

```
name = "Maya"
age = 25
height = 1.65
is_student = True
```

Common types: `str` (text), `int` (whole number), `float` (decimal), `bool` (True/False).

2. Input, output & conversion

```
print("Hello")
name = input("Your name: ")
age = int(input("Your age: "))
```

Useful conversions: `int()`, `float()`, `str()`, `bool()`.

3. Operators

Math: `+ - * / // % **`
Compare: `== != > < >= <=`
Logic: `and or not`

4. Decisions (if / elif / else)

```
mark = 72
if mark >= 90:
    print("A grade")
elif mark >= 50:
    print("Pass")
else:
    print("Try again")
```

5. Loops

for repeats over a sequence; **while** repeats while a condition is true.

```
for number in range(1, 4):
    print(number)

count = 1
while count <= 3:
    print(count)
    count += 1
```

6. Collections

List — ordered, changeable:

```
fruits = ["apple", "mango"]
fruits.append("grape")
print(fruits[0])
```

Tuple — ordered, unchangeable: `(10, 20)`
Set — unique values: `{1, 2, 3}`
Dictionary — key/value pairs:

```
person = {"name": "Maya", "age": 25}
print(person["name"])
```

7. Functions

A function is reusable code. Define it with `def`.

```
def greet(name):
    return "Hello, " + name

message = greet("Maya")
print(message)
```

8. Strings & useful methods

```
word = " python "
word.strip()    # remove edge spaces
word.upper()   # uppercase
word.lower()   # lowercase
len(word)       # length
```

9. Errors & files

```
try:
    number = int("abc")
except ValueError:
    print("Enter a valid number")
```

Read a text file safely:

```
with open("notes.txt", "r") as file:
    content = file.read()
```

10. Modules & imports

A module is reusable Python code. Import it to use its features.

```
import math
print(math.sqrt(25))
```

Quick rules to remember

- Indentation (usually 4 spaces) defines code blocks.
- Python is case-sensitive: `Name` and `name` differ.
- Use # for comments.
- = assigns; == compares.
- Lists use indexes starting at 0.

Learning tip: Change the example values, run the code, and predict the output before you execute it. Understanding the flow is more important than memorising syntax.